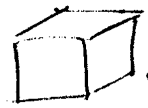
Total points : 35

PHY 632H

Solutions : HW # 2

Ans 10.2. Let $|E\rangle$ represent the state vector

$$H|E\rangle = E|E\rangle$$



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$$\therefore \left(\frac{P_x^2}{2m} + \frac{P_y^2}{2m} + \frac{P_z^2}{2m}\right)|E\rangle = E|E\rangle$$

In $x$ representation,

$$\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2} - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial z^2}\right)\psi_E(x,y,z) = E\,\psi_E(x,y,z)$$

$$\psi_E(x,y,z) = \psi_{E_1}(x)\,\psi_{E_2}(y)\,\psi_{E_3}(z) \quad \ldots \text{ Ansatz}$$

$$\therefore \psi_{E_2}(y)\,\psi_{E_3}(z)\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}\psi_{E_1}(x)\right) - \frac{\hbar^2}{2m}\psi_{E_1}(x)\,\psi_{E_3}(z)\frac{\partial^2\psi_{E_2}(y)}{\partial y^2}$$

$$-\frac{\hbar^2}{2m}\psi_{E_1}(x)\,\psi_{E_2}(y)\frac{\partial^2\psi_{E_3}(z)}{\partial z^2} = E\,\psi_{E_1}(x)\,\psi_{E_2}(y)\,\psi_{E_3}(z)$$

$$\therefore \underbrace{\frac{1}{\psi_{E_1}(x)}\left(-\frac{\hbar^2}{2m}\frac{\partial^2\psi_{E_1}}{\partial x^2}\right)}_{\text{depends only on } x} + \underbrace{\frac{1}{\psi_{E_2}(y)}\left(-\frac{\hbar^2}{2m}\frac{\partial^2\psi_{E_2}}{\partial y^2}\right)}_{\text{depends only on } y} + \underbrace{\frac{1}{\psi_{E_3}(z)}\left(-\frac{\hbar^2}{2m}\frac{\partial^2\psi_{E_3}}{\partial z^2}\right)}_{\text{depends only on } z} = E$$

$\therefore$ Each term is separately equal to a constant

$$\therefore -\frac{\hbar^2}{2m}\frac{\partial^2\psi_{E_1}}{\partial x^2} = E_1\,\psi_{E_1}(x) \;;\; -\frac{\hbar^2}{2m}\frac{\partial^2\psi_{E_2}(y)}{\partial y^2} = E_2\,\psi_{E_2}(y) \ldots$$

$$-\frac{\hbar^2}{2m}\frac{\partial^2 \Psi_{E_1}}{\partial x^2} = E_1 \Psi_{E_1}(x) \quad \& \quad \Psi_{E_1}(x) = 0 \quad \text{if } x<0 \text{ or } x>L$$

→ we know the solution for this is

$$\Psi_{E_1}(x) = \sqrt{\frac{2}{L}}\sin\left(\frac{n_x \pi x}{L}\right) \quad \& \quad E_1 = \frac{n_x^2 \pi^2 \hbar^2}{2mL^2}$$

|||ly $\Psi_{E_2}(y) = \sqrt{\frac{2}{L}}\sin\left(\frac{n_y \pi y}{L}\right)$ ; $E_2 = \frac{n_y^2 \hbar^2 \pi^2}{2mL^2}$

& $\Psi_{E_3}(z) = \sqrt{\frac{2}{L}}\sin\left(\frac{n_z \pi z}{L}\right)$ , $E_3 = \frac{n_z^2 \hbar^2 \pi^2}{2mL^2}$

By our ansatz, $\Psi_E(x,y,z) = \Psi_{E_1}(x)\,\Psi_{E_2}(y)\,\Psi_{E_3}(z)$

$$\therefore \Psi_E(x,y,z) = \sqrt{\frac{2}{L}}\sin\left(\frac{n_x \pi x}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n_y \pi y}{L}\right)\sqrt{\frac{2}{L}}\sin\left(\frac{n_z \pi z}{L}\right)$$

$$n_x, n_y, n_z = 1, 2, 3, \ldots$$

$$\& \; E = E_1 + E_2 + E_3$$

$$= \frac{\hbar^2 \pi^2}{2mL^2}\left(n_x^2 + n_y^2 + n_z^2\right)$$

Ans 10.2.2 $\mathcal{H} = \frac{P_x^2}{2m} + \frac{P_y^2}{2m} + \frac{1}{2}m\omega_x^2 x^2 + \frac{1}{2}m\omega_y^2 y^2$

$$H|\Psi\rangle = E|\Psi\rangle$$

In $x$ representation,

$$\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2 - \frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2} + \frac{1}{2}m\omega^2 y^2\right)\Psi(x,y) = E\Psi(x,y)$$

Ansatz : $\Psi(x,y) = \Psi_x(x)\,\Psi_y(y)$

$$\therefore \underbrace{\frac{1}{\psi_x(x)}\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}+\frac{1}{2}m\omega_x^2x^2\right)\psi_x}_{E_x} + \underbrace{\frac{1}{\psi_y(y)}\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial y^2}+\frac{1}{2}m\omega_y^2y^2\right)\psi_y}_{E_y} = E$$

$(\therefore E_x + E_y = E)$

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi_x}{\partial x^2}+\frac{1}{2}m\omega_x^2x^2\psi_x = E_x\psi_x$$

$$\& \quad -\frac{\hbar^2}{2m}\frac{\partial^2\psi_y}{\partial y^2}+\frac{1}{2}m\omega_y^2y^2\psi_y = E_y\psi_y.$$

We know from Sec 7.3, Shankar that equations as above will admit a solution only if,

$$E_x = (n_x+\tfrac{1}{2})\hbar\omega, \quad n_x = 0,1,2\ldots$$

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$$\& \quad E_y = (n_y+\tfrac{1}{2})\hbar\omega, \quad n_y = 0,1,2,\ldots$$

$$\therefore E = (n_x+\tfrac{1}{2})\hbar\omega + (n_y+\tfrac{1}{2})\hbar\omega; \quad n_x, n_y = 0,1,2,\ldots$$

(2) $\psi_E(x,y) = \psi_E(x)\,\psi_y(y)$

$$= \left(\frac{m\omega_x}{\pi\hbar\, 2^{2n_x}(n_x!)^2}\right)^{1/4}\left(\frac{m\omega_y}{\pi\hbar\, 2^{2n_y}(n_y!)^2}\right)^{1/4}\exp\left(-\frac{m\omega_x x^2}{2\hbar}\right)\exp\left(-\frac{m\omega_y y^2}{2\hbar}\right)$$

$$H_{n_x}\left[\left(\frac{m\omega_x}{\hbar}\right)^{1/2}x\right] H_{n_y}\left[\left(\frac{m\omega_y}{\hbar}\right)^{1/2}y\right]$$

$$H_{n_x}\left[\left(\frac{m\omega_x}{\hbar}\right)^{1/2}(-x)\right] = (-1)^{n_x} H_{n_x}\left[\left(\frac{m\omega_x}{\hbar}\right)^{1/2}x\right]$$

$$H_{n_y}\left[\left(\frac{m\omega_y}{\hbar}\right)^{1/2}(-y)\right] = (-1)^{n_y} H_{n_y}\left[\left(\frac{m\omega_y}{\hbar}\right)^{1/2}y\right]$$

Thus $\psi_{E_n}(-x,-y) = (-1)^{n_x+n_y}\ \psi_{E_n}(x,y)$

$= -\psi_{E_n}(x,y)$ if $n$ odd

$= \psi_{E_n}(x,y)$ if $n$ even.

(3) $\psi_{n=0}(x,y) = \left(\frac{m\omega_x}{\pi\hbar\, 1}\right)^{1/4}\left(\frac{m\omega_x}{\pi\hbar}\right)^{1/4} exp\left(\frac{-m\omega x^2}{2\hbar}\right) exp\left(\frac{-m\omega y^2}{2\hbar}\right) 1\ 1$

$(\omega_x = \omega_y = \omega)$

$\therefore \psi_{n=0}(x,y) = \sqrt{\frac{m\omega}{\pi\hbar}}\ exp\left(-\frac{m\omega}{2\hbar}(x^2+y^2)\right)$

$\therefore \psi_{n=0}(\rho,\phi) = \sqrt{\frac{m\omega}{\pi\hbar}}\ exp\left(\frac{-m\omega}{2\hbar}\rho^2\right)$

$\psi_{n=1\ (0,1)}\overset{x,y}{(\rho,\phi)} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\left(\frac{m\omega}{\pi\hbar\, 2^2}\right)^{1/4} exp\left(-\frac{m\omega}{2\hbar}(x^2+y^2)\right) 1\ (2y)\left(\frac{m\omega}{\hbar}\right)^{1/2}$

$\psi_{n=1\ (0,1)}(\rho,\phi) = \sqrt{\frac{m\omega}{2\pi\hbar}}\ exp\left(-\frac{m\omega}{2\hbar}\rho^2\right) 2\rho\sin\phi\sqrt{\frac{m\omega}{\hbar}}$

$\psi_{n=1\ (n_x=1,\, n_y=0)}(x,y) = \left(\frac{m\omega}{2\pi\hbar}\right)^{1/2} exp\left(-\frac{m\omega}{2\hbar}(x^2+y^2)\right) 2x \left(\frac{m\omega}{\hbar}\right)^{1/2}$

$\therefore \psi_{n=1}(\rho,\phi) = \frac{1}{\sqrt{2\pi}}\left(\frac{m\omega}{\hbar}\right) exp\left(\frac{-m\omega}{2\hbar}\rho^2\right) 2\rho\cos\phi$

$E = (n+1)\hbar\omega = (n_x + 1/2 + n_y + 1/2)\hbar\omega \quad ; \quad n_x + n_y = n$

| | | $n_x$ | $n_y$ |
|---|---|---|---|
| | Total | 0 | $n$ |
| ~~degene~~ degeneracy | possible | 1 | $n-1$ |
| is $(n+1)$ | ways $(n+1)$ | 2 | $n-2$ |
| | | $\vdots$ | |
| | | $n$ | 0 |

Ans 10.3.2 $|\psi\rangle = N\left[|3,3,4\rangle + |3,4,3\rangle + |4,3,3\rangle\right]$

Using $\langle\psi|\psi\rangle = N$ ; $N = \frac{1}{\sqrt{3}}$

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$\therefore |\psi\rangle = \frac{1}{\sqrt{3}}\left[|3,3,4\rangle + |3,4,3\rangle + |4,3,3\rangle\right]$

Ans 10.3.3

1, 2, 3 — c, b, a

particles are labelled

→ allowed distinct configurations

the possible choices for each particle are 3.

$\therefore$ Total number of possible configurations $= 3 \times 3 \times 3 = \boxed{27}$

(2) Bosons are indistinguishable

possible distinct configurations are $\boxed{10}$

1) a a a
2) a a b
3) a a c
4) b b b
5) b b a
6) b b c
7) c c a
8) c c b
9) c c c
10) a b c

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(3) Fermions are indistinguishable & no two fermions can occupy the same energy level. So there is only one possible configuration, abc.

Ans 10.3.4 For a particle in box, $E_n = \frac{n^2 \hbar^2 \pi^2}{2mL^2}$

$E_{system} = \frac{\pi^2 \hbar^2}{mL^2} \Rightarrow E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$ & $E_2 = \frac{\pi^2 \hbar^2}{2mL^2}$

$\therefore$ both particles have same energy, they have to be bosons.

$\therefore |\psi\rangle = |E_1\rangle |E_1\rangle$

$$E_{system} = \frac{5\hbar^2\pi^2}{2mL^2} = \frac{4\pi^2\hbar^2}{2mL^2} + \frac{\hbar^2\pi^2}{2mL^2}$$

$\therefore$ Possible states, $|\psi\rangle = \frac{1}{\sqrt{2}}\left[|E_1\rangle|E_2\rangle + |E_2\rangle|E_1\rangle\right]$ ... symmetric

or $|\psi\rangle = \frac{1}{\sqrt{2}}\left[|E_1\rangle|E_2\rangle - |E_2\rangle|E_1\rangle\right]$ ... antisymmetric.

Ans 10.3.1 $|S\rangle = N\left[|\phi,\psi\rangle + |\psi,\phi\rangle\right] = N\left[|\phi\rangle\otimes|\psi\rangle + |\psi\rangle\otimes|\phi\rangle\right]$

(3) $\therefore N^2\left(\langle\phi|\otimes\langle\psi| + \langle\psi|\otimes\langle\phi|\right)\left(|\phi\rangle\otimes|\psi\rangle + |\psi\rangle\otimes|\phi\rangle\right) = 1$

$$\therefore \frac{1}{N^2} = \langle\phi|\phi\rangle\langle\psi|\psi\rangle + \langle\phi|\psi\rangle\langle\psi|\phi\rangle + \langle\psi|\phi\rangle\langle\phi|\psi\rangle + \langle\psi|\psi\rangle\langle\phi|\phi\rangle$$

$$= 2\langle\phi|\phi\rangle\langle\psi|\psi\rangle + 2\langle\psi|\phi\rangle\langle\phi|\psi\rangle$$

$$= 2 + 2\,|\langle\phi|\psi\rangle|^2$$ ... (assuming $|\phi\rangle$ & $|\psi\rangle$ are normalized)

$$\therefore |S\rangle = \frac{1}{\left[2 + 2|\langle\phi|\psi\rangle|^2\right]^{1/2}}\left[|\phi\rangle\otimes|\psi\rangle + |\psi\rangle\otimes|\phi\rangle\right]$$

Ans 10.3.6 Fermions have an half integer spin $n\frac{\hbar}{2}$, $n$ odd & Bosons have n integral spins, $n\hbar$.

If we have even number of fermions, the total spin will add up to an integers & thus the particle will behave as a boson.

If we have odd number of fermions, the total spin will add up to an half integer & the particle will behave as a fermion.

H atom has a proton (fermion) & an electron (also a fermion). So it is boson.